Modeling Climate Change Over Time

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Introduction

- So, why should we care?

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Data

Does the presentation address the sources, reliability, and suitability of their data for the problems they are addressing?

Data - Volcanoes

- Is global climate change due primarily to volcanoes?

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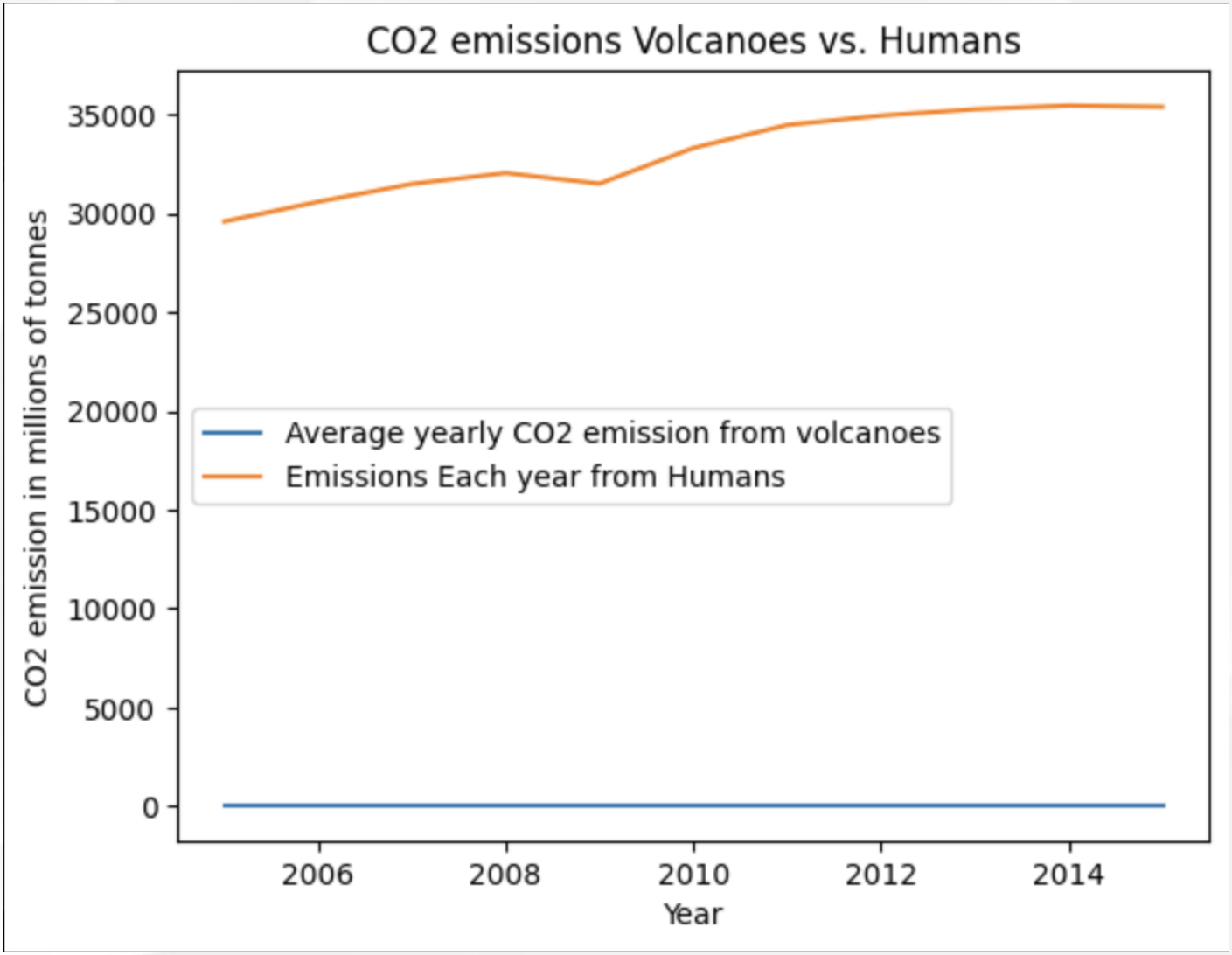


Data - A Falsified Dataset

- One Kaggle dataset claimed to use data from a reputable source

About this file

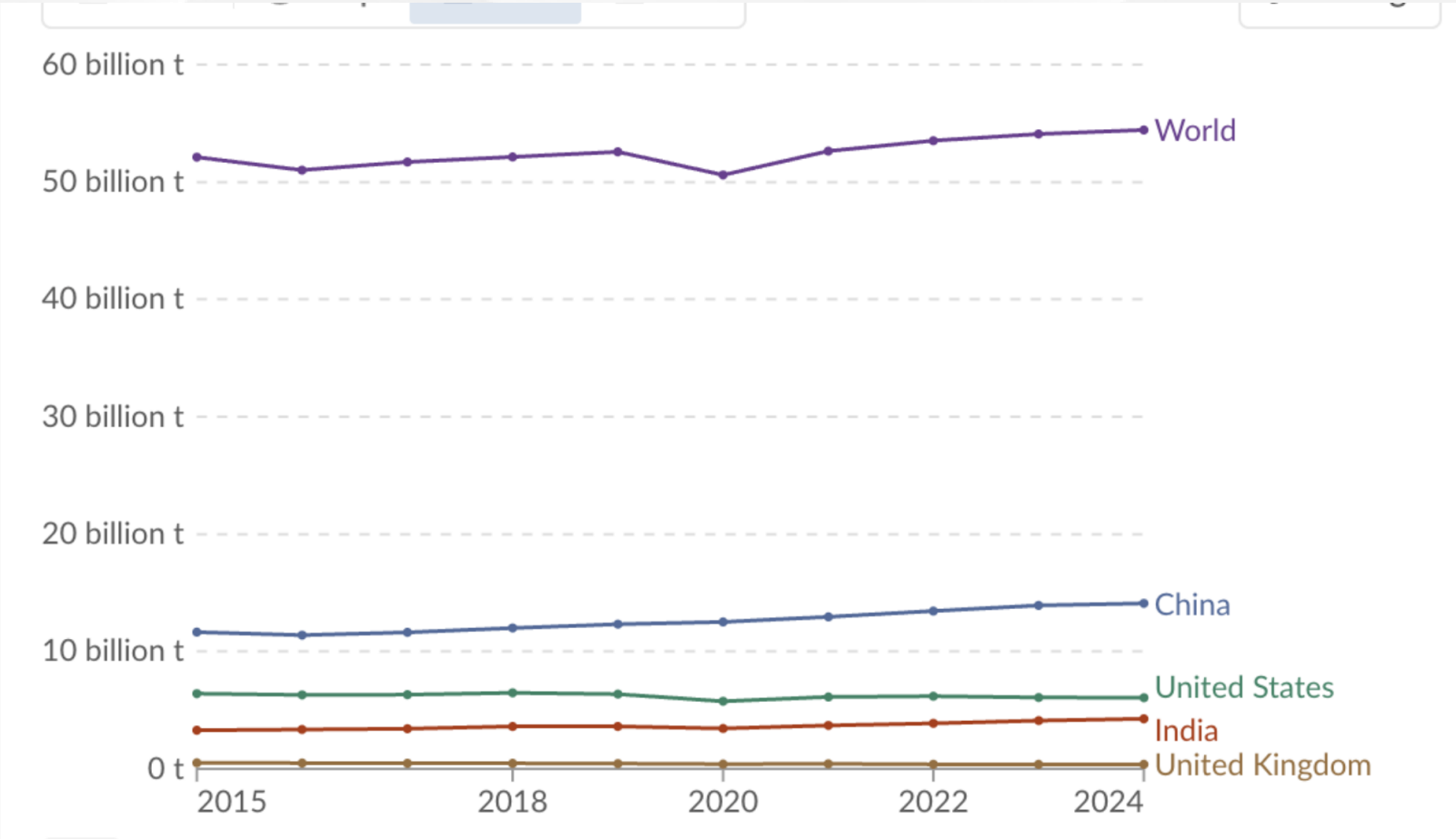
A csv with country and years as column.
Each row shows a country and its emission data from 1750 -2019 measured in tonnes.
Country column is String, while other columns are integer

Country Country Names	# 2017
231 unique values	 0 1580b
World	1.58E+12
United States	3.99E+11
EU-28	3.53E+11
China	2.00E+11
Asia and Pacific (other)	1.78E+11
Europe (other)	1.58E+11
Russia	1.01E+11

Data - A Falsified Dataset

BYU

- A graph from the actual source:



Methods

We used many techniques:

- Mixture models
  - Gaussian & linear
- Kalman filter
- Seasonal time series decomposition
- ARIMA
  - May mix up trends and seasonal behavior
  - SARIMA incorporates seasonal decompositions

GMM-Fit Regression

Has the course of temperatures in the United States fundamentally changed in behavior?

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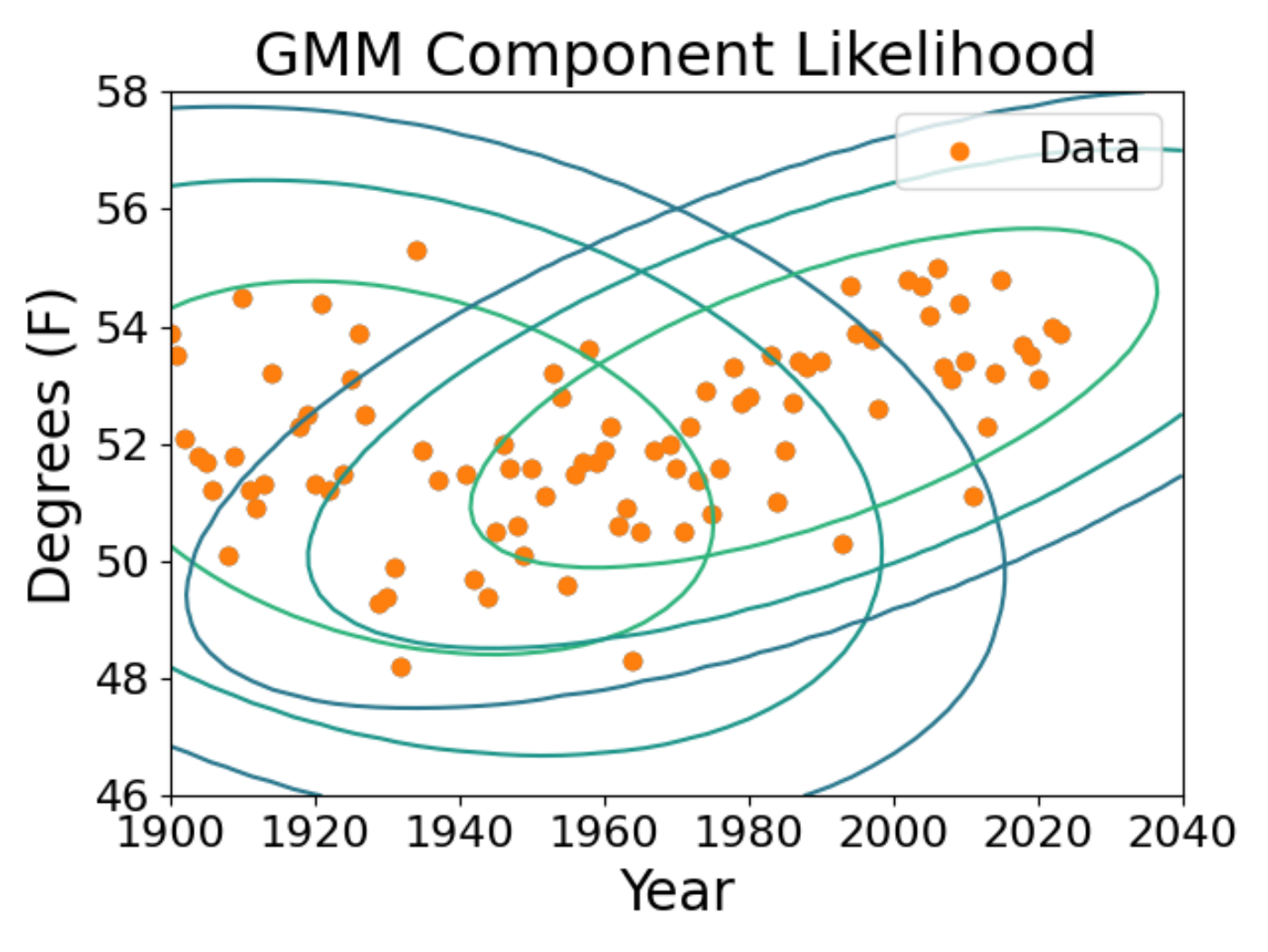
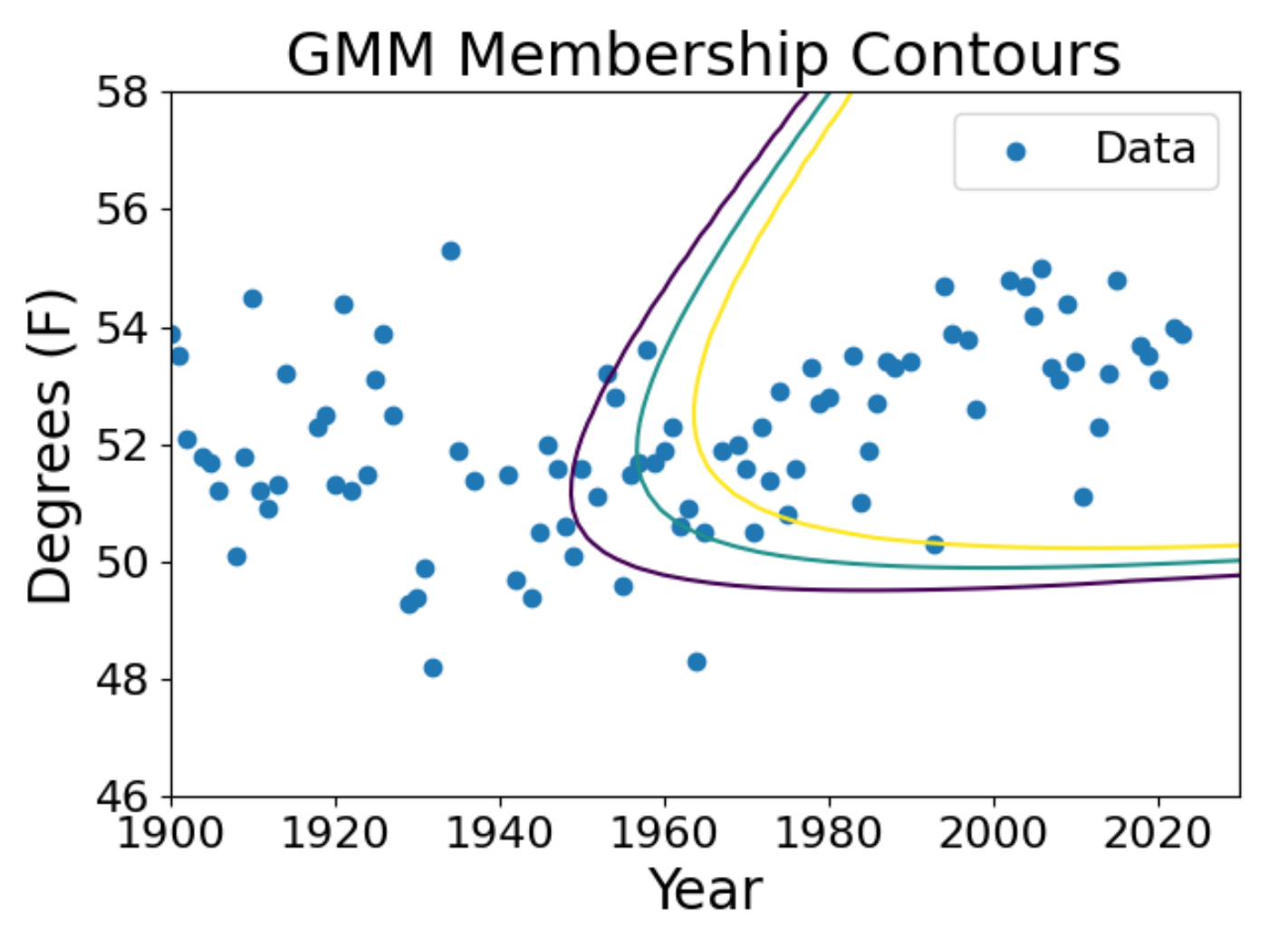
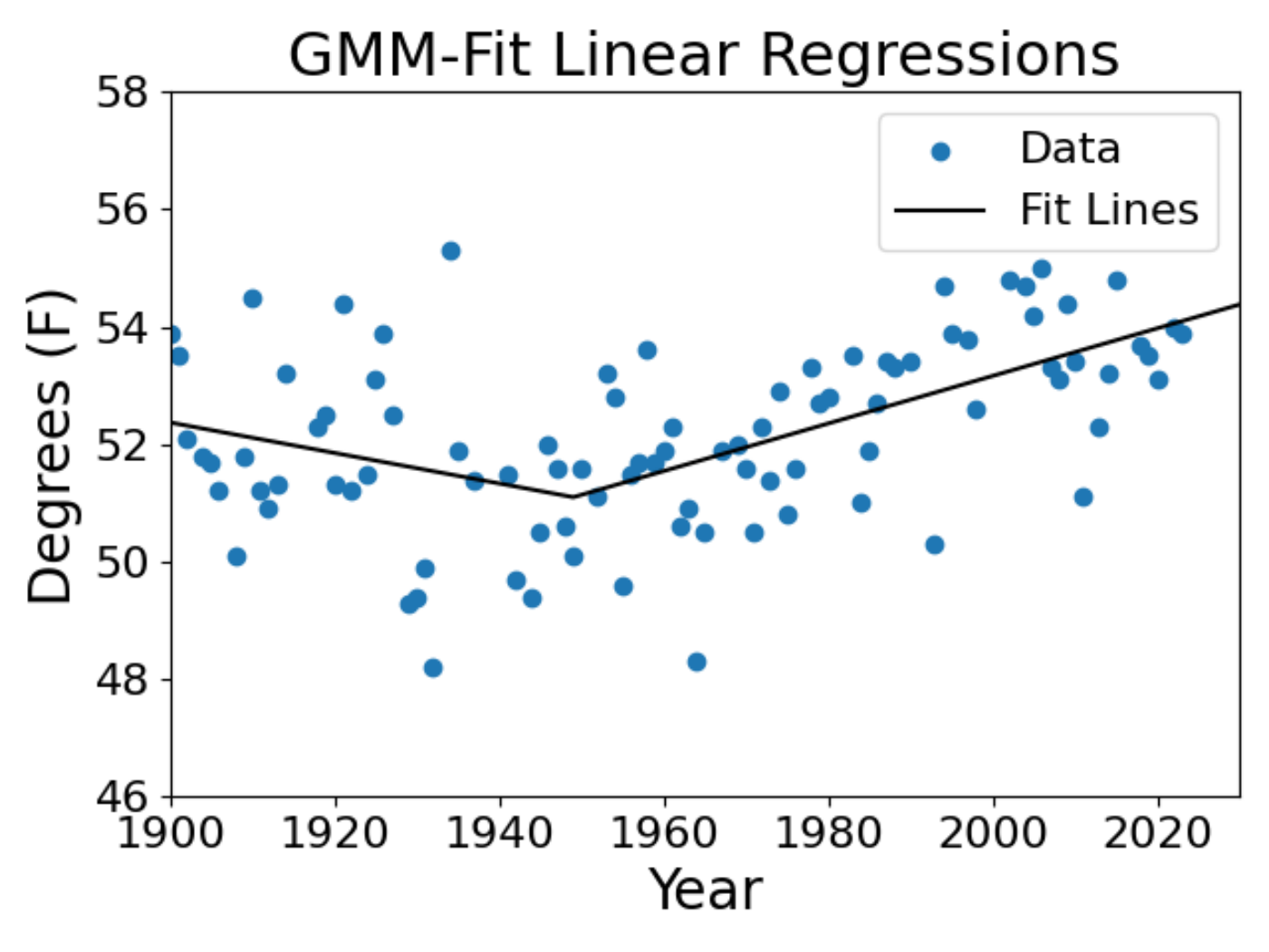


GMM-Fit Regression, Step 2



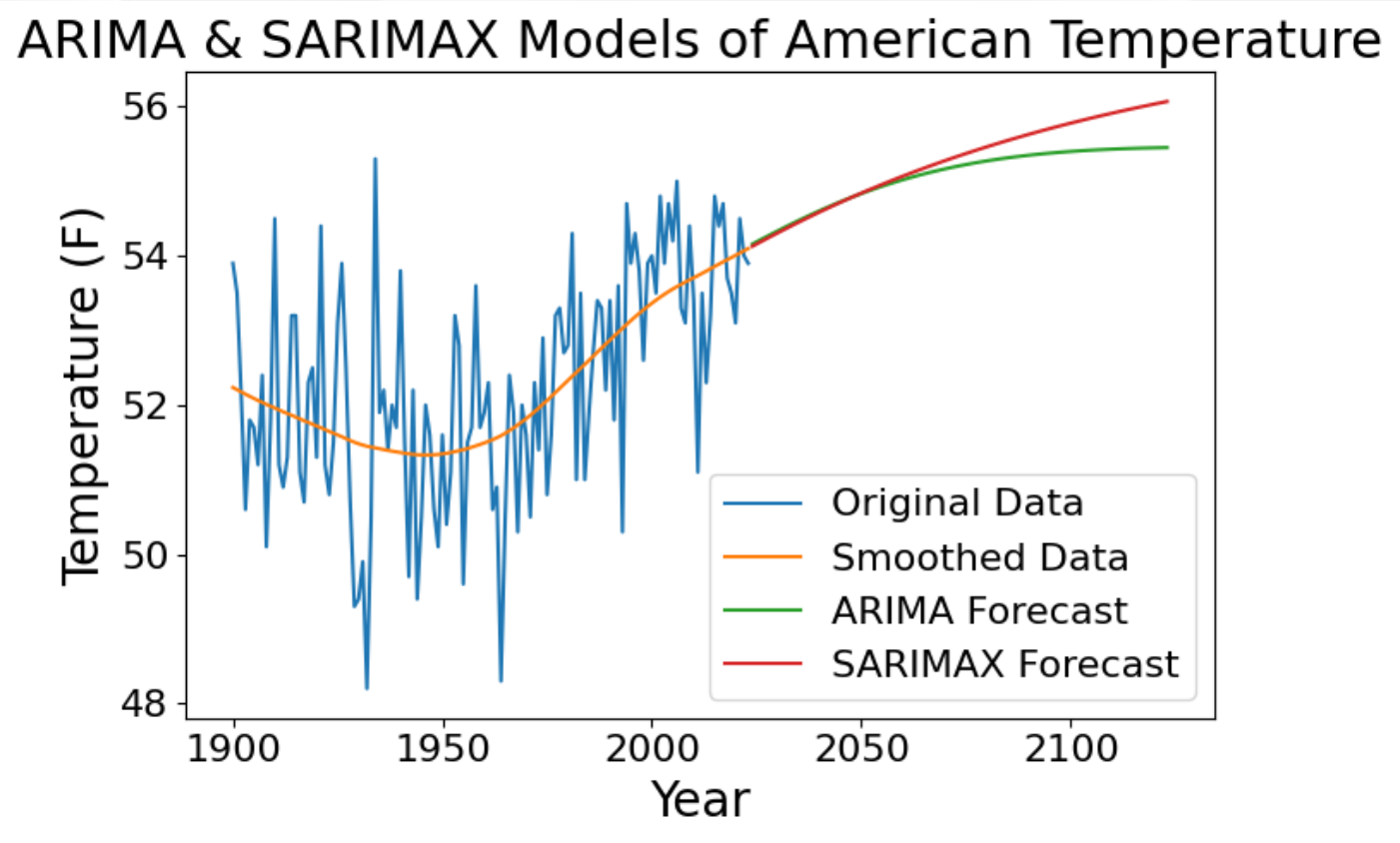
GMM-Fit Regression, Step 3



Results and Analysis - Nathan

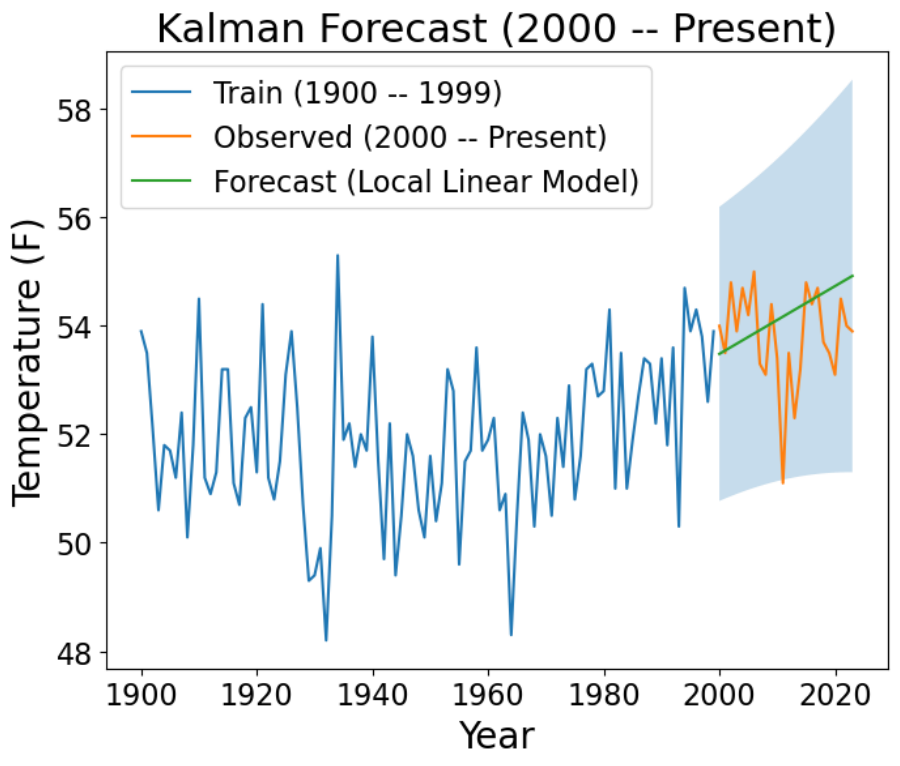
Do they clearly and succinctly describe the results and their interpretation and analysis of the results? Does the presentation demonstrate a thorough understanding of the potential challenges and weaknesses of the methods and models? Do they clearly summarize the conclusions and outstanding open questions? Are the results significant or meaningful?

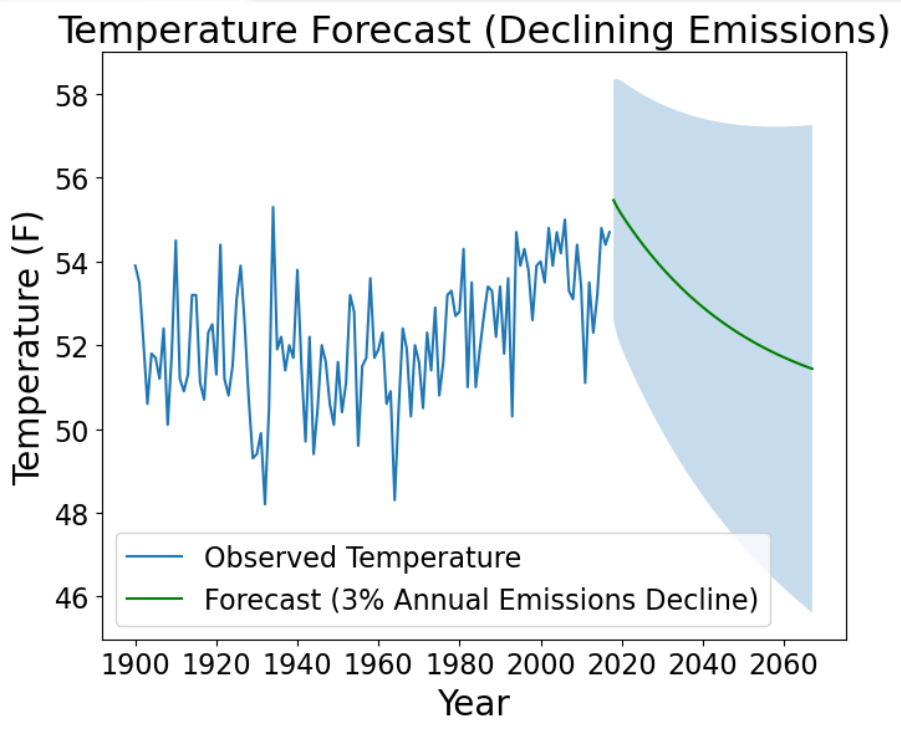
Results and Analysis - SARIMA



Kalman Filter Predictions: Good Cop/Bad Cop

- ARIMA modeling with Kalman filter
- Current trajectory vs. 3% emissions decline





Ethics

- Collecting data: **ethical**
- Ignorance of what data indicates: **not ethical**
- Our results do not assign culpability to any individual person, industry or nation
  - Propaganda about global climate change is real
- Everyone should hold themselves accountable

Thank you

Thank you for your attention